MIL-S-80001C
20 January 1984
SUPERSEDING
MIL-S-80001B
18 August 1978

MILITARY SPECIFICATION

SHAPERS, METAL CUTTING, VERTICAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers vertical metal cutting shapers or slotters, either mechanically or hydraulically driven, suitable for accurate and rapid metal removal,

1.2 Classification. This specification covers vertical shapers or slotters of the following types and sizes. The type and size machine required shall be as specified (see 6.2.1).

Type I - Hydraulically Driven

Size	Ram stroke (inches)
1	12
2	20
3	36

Type II - Mechanically Driven

Size	Ram stroke (inches)
1	12
2	18
3	24
4	30

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, Memphis, Tennessee 38114, by using the self addressed standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3418

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards and handbooks of the issue listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standard for Federal Services

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

U. S. DEPARTMENT OF LABOR

OSHA 2206 - General Industry, OSHA Safety and Health Standards (29 CFR1910)

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Officer, Washington, DC 20402.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

AMERCIAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/NFPA No. 79 - Metalworking Machine Tools and Plastic Machinery, Electrical Standard for
ANSI/B5.1 - T-Slots, Their Bolts, Nuts, Tongues, and Cutters
ANSI/IEEE 268 - Metric Practices

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(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM-D-3951 - Commercial packaging, Standard Practice for

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

MG1 - Motors and Generators
ICS - Industrial Controls and Systems

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street, NW, Washington, DC 20037.)

2.3 Order of precedence. In the event of conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS .

3.1 First article. When specified (see 6.2.1), the contractor shall furnish one complete machine of each type and size specified for first article inspection and approval (see 4.3 and 6.3).

3.2 Design. The machine shall be new and one of the manufacturer's current models capable of shaping metal in accordance with the requirements herein. The machine shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2.1 Measurement systems. Unless otherwise specified, either the U. S. Customary System of Units (US) or the International System of Units (SI) may be used in the design and construction of the machine. When only one system of measurements is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all measurements, dimensions, sizes and capacities are given in the U. S. Customary System of Units (US). These measurements may be converted to the International System of limits (SI) through the use of the conversion factors and methods specified in ANSI/IEEE 268.

3.2.1.1 Measuring and indicating device calibrations. When specified (see 6.2.1), measuring and indicating devices such as feed screw dials, scales, depth stops, carriage stops, dial indicators, pressure gauges, temperatures indicators, and all other similar devices shall be graduated in the specified system (US or SI) of measurements. Regardless of the

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measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system.

3.2.1.2 Dual calibrations. When specified (see 6.2.1), measuring and indicating devices shall be graduated in both the US and the SI system of measurements. When a dual US and SI system is furnished, both feed dials shall have independent zero adjustments and both dials shall be calibrated in such a manner that the last dial graduation progresses into and is continuous with the first dial graduation as the dial is rotated through the zero position.

3.2.2 Reclaimed materials. The machine may contain reclaimed materials Provided such materials will not jeopardize the machine's intended use and performance. The reclaimed materials shall have been reprocessed, remanufactured or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the machine.

3.2.3 Controls. All operating controls shall be located convenient to the operator at his normal work station.

3.2.4 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the machine that present safety hazards. The safety devices shall-not interfere with operation of the machine. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance and repair of the parts. All machine parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall comply with all of the requirements of OSHA 2206 that are applicable to the machine itself. *Additional safety and health requirements shall be as specified (see 6.2.1 and 6.4).

3.2.5 Lubrication. Means shall be provided to insure adequate lubrication for all moving parts. Each recirculating system shall include a filter which is cleanable or replaceable. Each lubricant reservoir shall have means for determining fluid level. All oil holes, grease fittings and filler caps shall be accessible.

3.2.6 Energy efficiency. The shaper machines and all applicable components that directly consume energy in normal operation shall be designed and constructed for maximum energy efficiency as governed by the latest developments available within the industry.

3.3 Construction. The machine shall be constructed of parts which are new, without defects and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free of scale and mismatching. No process such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective parts.

3.3.2 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.3.3 Fastening devices. All screws, pins, bolts and other fasteners shall be installed in a manner that prevents change of tightness. Those subject to removal or adjustment shall not be swaged, peened, staked or otherwise permanently installed.

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3.3.4 Surfaces. All surfaces shall be cleaned and free of sand, dirt, fins, sprues, flash, scale, flux and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.5 Painting. Unless otherwise specified (see 6.2.1), painting shall be in accordance with the manufacturer's commercial practice.

3.3.6 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable *'Detailed Standard" section referenced therein.

3.3.7 Dials. All dials that indicate stock removal or tool movement shall be graduated in increments not larger than .001 inch. Dial diameters shall be such that graduations may be read from the normal operating position. Dials shall be permanently and legibly engraved or etched on a nonglare background.

3.4 Components.

3.4.1 Coolant system. The coolant system shall include a sump or reservoir; a power driven pump and all necessary piping, and shall have sufficient capacity to permit full flow of coolant. The system shall have means for draining and cleaning and shall include a strainer or baffle which is easily removable for cleaning. Means shall be provided to permit the operator to direct and control the amount of coolant.

3.4.2 Hydraulic system. The hydraulic system, when furnished, shall be complete, including all pumps, valves, piping, cylinders, and pressure controls. overpressure protection shall be provided in the high pressure line to prevent damage to components. A filter system shall be provided to insure delivery of clean fluid. The hydraulic reservoir shall have means for determining fluid.

3.4.3 Electrical system. Unless otherwise specified (see 6.2.1), the electrical system shall conform to all requirements in these paragraphs. The system shall conform to the requirements of NFPA-79. Each machine shall draw all of its electrical power from a single 230/460 volt, 3 phase, 60 Hz circuit having a fused safety disconnect switch or circuit breaker. An identified terminal for-grounding the machine when installed shall be mounted in or near the disconnect switch. The terminal shall be suitable for connecting the size grounding conductor specified in table 240-A, NFPA-790

3.4.3.1 Motors. Motors shall have ball or roller bearings. Unless otherwise specified (see 6.2.1), each motor shall meet the requirements of a dripproof enclosure. Motors shall conform to NEMA Standard MG-1.

3.4.4 Bed and column. The bed and column shall be either a casting of impact resistant iron or a stress relieved fabricated steel weldment designed for floor mounting. The structure shall be internally ribbed and braced to minimize distortion and deflection. Doors or readily removable covers shall be provided to permit servicing the drive motor and other internal mechanisms.

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,3.4.5 Ram. The ram shall be impact resistant iron or iron-alloy of the dovetail slide or square lock construction. An adjustable gib for maintaining alignment shall be provided and shall have full bearing throughout the length of the ram. Means for adjusting the stroke length shall be provided. Stroke length shall conform to tables I and 11. Horizontal angular adjustment for the ram guide shall be provided. Means for clamping the ram guide in any desired position of angular adjustment shall be provided. The ram shall be positioned vertically by means of a hand crank and screw or equal device. Means shall be provided for adjusting and locking the ram stroke at any length within the capacity of the machine.

3.4.6 Ram angle. It shall be possible to position the ram guide from zero degrees to not less than ten degrees forward from the true vertical. A positive stop shall be provided to permit the ram guide to be returned to a true vertical position without realignment. The ram guide shall have an indicator scale permanently and legibly marked in degrees, with fifteen minute subgraduation.

3.4.7 Ram drive. The reciprocating action of the ram shall be obtained by either a type I, hydraulically driven, or type 11, mechanically driven, drive as specified (see 1.2 and 6.2.1). If a hydraulic drive is used, provisions shall be made to equalize back pressures and cutting pressures to assure a uniform speed when taking interrupted cuts, to produce a finish which is free from undesirable machining marks. It shall be possible to stop and hold the ram in any desired position. Both the hydraulic and the mechanical drives shall be such that the return speed of the ram is 25 percent greater than the cutting speed. An overload protection device shall be provided on the ram drive to prevent damage should the ram be driven into the vise or other obstruction, and in cases of over shooting the original settings.

3.4.8 Ram head. The ram head shall be of impact resistant iron or iron-alloy and shall be designed to resist thrust without perceptible vibration and deflection resulting from heavy cuts. The head shall be provided with a clapper box capable of swiveling at least 180 degrees. The clapper box shall be machined to receive a single post tool holder, and shall be provided with means to lift the tool on the return stroke. The tool post shall be a steel forging with an opening to receive tools of the size specified in tables I and II. Lock screws shall be provided to make the tool rigid when performing keywaying operations.

3.4.9 Saddle. The saddle shall be a one piece casting of impact resistant iron alloy or a stress relieved steel fabrication designed and proportioned to withstand heavy cuts. The saddle shall be provided with hand feeds, power feeds and power rapid traverse in two directions, both longitudinal and traverse. The saddle slide shall have adjustable gibs the full bearing length of the saddle way. The saddle ways shall be provided with covered wipers.

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3.4.10 Rotary table and dividing head. The table shall be a circular casting of impact resistant iron or iron alloy. The table shall be capable of swiveling 360 degrees, and shall be marked permanently in degrees with 15 minutes subgraduations. The table shall be equipped with a built-in dividing head for accurate spacing of keyways. The top of the table shall be provided with not less than four T-slots in accordance with ANSI B5.1. One T-slot shall run through the center of the table perpendicular to the remaining three or more T-slots, which shall be parallel with each other and equally spaced. The table slide shall have adjustable gibs the full bearing length of the table.

3.4.11 Table feed. The table feeds shall conform to table I or II. The amount of feed, as well as direction of feed, shall be regulated through controls convenient to the operator. Hand feeds, power feeds, and power rapid traverse shall be provided in all directions: longitudinal, transverse and rotary. A suitable device shall be incorporated to disengage all feeds when they have reached a predetermined limit of travel in any direction.

3.4.12 Power rapid traverse. Table slide, saddle, and rotary movements shall have power feed, and shall be provided with power rapid traverse in all directions: longitudinal, transverse, and rotary. Automatic disengagement of the normal feeds shall occur when rapid traverse is engaged.

3.4.13 Feed screw. The longitudinal and transverse feed screws shall be of ball or roller type bearings. If a stationary screw and rotating nut design is used, the rotating nut only shall be mounted on ball or roller type bearings. The screws shall be fitted with direct reading micrometer dials or digital type readout graduated in increments of 0.001 inch. Provisions shall be made to permit adjusting and locking the feed screws in any desired position. Adequate means of lubrication for the feed screws shall be provided.

3.5 Sizes and capacities. The sizes and capacities of the machine shall be not less than the requirement of tables I and II.

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TABLE I - Type I Hydraulically Driven, Sizes and Capacities (see 3.5)

CHARACTERISTICS (Linear Dimensions in Inches Unless Otherwise Indicated)	Size 1 12"	Size 2 20"	Size 3 36"
Stroke of ram			
Minimum, not more than	3	3	3
Maximum, not less than	13	22	40
Cutting strokes or ram, Feet Per Minute (FPM)			
Low, not more than	25	25	10
High, not less than	100	100	100
Table top to end of ram, longest stroke			
Minimum, not more than	1-1/4	1-1/4	1-1/4
Maximum, not less than	33	42	64
Distance column face to center of table			
Minimum, not more than	19-1/4	19-1/4	19-1/4
Maximum, not less than	50	50	50
Vertical adjustment of ram, not less than	22	22	22
Rotary table diameter, not less than	28	28	28
Longitudinal travel of table, not less than	32	32	32
Transverse travel of table, not less than	24	24	24
Angular adjustment of ram, degrees, range	0 to 10	0 to 10	0 to 10
Feeds, per stroke, longitudinal and travel			
Minimum, not more than	0.005	0.005	0.005
Maximum, not less than	0.125	0.125	0.125
Rotary power feed, per stroke (on-inch radius)			
Minimum, not more than	0.005	0.005	0.005
Maximum, not less than	0.200	0.200	0.200
Tool post holder capacity, not less than	7/8 x 2	7/8 x 2	7/8 x 2
Table T-slot width, not less than	13/16	13/16	13/16
Horsepower of drive motor, not less than	10	10	20

Requirements are "not less than" those shown. Where a range is shown, the required performance is from the "stated minimum or less" to the "stated maximum or greater."

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TABLE II - Type II Mechanically Driven, Sizes and Capacities (see 3.5)

CHARACTERISTICS (Linear Dimensions in Inches Unless Otherwise Indicated)	Size 1 12 "	Size 2 18"	Size 3 24"	Size 4 30"
Stroke of ram				
Minimum, not more than	3	3	3	3
Maximum, not less than	13	20	28	34
Cutting strokes or ram, Feet Per Minute (FPM)				
Low, not more than	5	5	5	5
High, not less than	100	100	100	100
Table top to end of ram, longest stroke				
Minimum, not more than	1-1/4"	1-1/4"	1-1/4"	1-1/4"
Maximum, not less than	28"	28"	43"	50"
Column face to tool post seat	26"	26"	26"	30"
Vertical adjustment of ram slide	7"	7"	7"	7"
Rotary table diameter not less than	28"	28"	36"	36"
Longitudinal travel of table	38"	38"	38"	52"
Transverse travel of table	25"	25"	25"	33"
Feeds, per stroke, longitudinal and travel				
Minimum, not more than	0.001	0.001	0.001	0.001
Maximum, not less than	.250	.250	.250	.250
Rotary powerfeed, per stroke (on 10-inch radius)				
Minimum, not more than	.0018	.0018	.0018	.0018
Maximum, not less than	.3600	.3600	.3600	.3600
Tool post holder capacity, not less than	1" x 2"	1" x 2"	1" x 2"	1" x 2"
Table T-slot width, not less than	13/16"	13/16	13/16	13/16
Horsepower of drive motor, not less than	7.5	10	15	20

Requirements are "not less than" those shown. Where a range is shown, the required performance is from the "stated minimum or less" to the "stated maximum or greater."

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3.6 Performance. Each machine shall be capable of producing cutting forces at maximum table cutting speed in accordance with tables I and 11.

3.7 Alignment tolerances. Each machine shall conform to the alignment tolerances specified in table III.

TABLE III - Alignment tolerances. (see 3.7)

REQUIREMENTS (All Dimensions In Inches)	Tolerances
Bedways, for longitudinal slide straight in longitudinal direction	0.0005 per 12
Bedways, for longitudinal slide flat or level in transverse direction	0.0005 per 12
Guides for transverse slides straight in direction at right angles to bedways	0.0005 per 12
Guides for transverse slide flat or level in direction of bedways	0.0005 per 12
Top of rotary table for concavity (no convex condition)	0.001
Incline of top of table transversely (maximum travel)	0.001
Incline of top of table longitudinally (maximum travel)	0.001
Rotate top of rotary table, indicator on an 18 inch diameter circle with indicator in tool post	0.002 TIR
Concentricity of center hole in rotary table	0.001 TIR
Alignment of ram slide with back and side of test square on work table (maximum stroke) (lower end toward column)	0.001
Cross slide perpendicular to longitudinal slide	0.0008 per 12
T-slots parallel with transverse motion of table in zero position	0.0008 per 12
T-slots parallel with longitudinal motion of table in the 90 degree position	0.0008 per 12

3.8 Standard equipment. Unless otherwise specified (see 6.2.1), the following equipment shall be furnished with each machine:

a. One set of hand tools normally furnished by the manufacturer.

3.9 Optional equipment. Optional equipment shall be furnished as specified and fully described (see 6.2.1).

3.10 Lubrication chart or plate. Unless otherwise specified (see 6.2.1), a lubrication chart or plate shall be permanently and securely attached to each machine. If a chart is furnished, it shall be placed in a transparent plastic folder, or permanently sealed between clear plastic sheets, with suitable means of mounting. The chart or plate shall-contain the following information:

Points of lubricant application
Servicing interval
Type of lubricant
Viscosity

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3.11 Nameplate, Unless otherwise specified (see 6.2.1), a corrosion-resistant metal nameplate shall be securely attached to each machine. The nameplate shall contain the information listed below. If the machine is a special model, the model designation shall include the model of the basic standard machine and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated provided the entry for each caption is clear as to its identity.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Contract Number of Order Number
National Stock Number or Plant Equipment Code
Date of Manufacture

3.12 Technical data. Technical data shall be furnished as specified (see 6.2,2),

3.13 Workmanship. Workmanship of the machine and accessories shall be of a quality equal to that of the manufacturer's commerical equipment of the type specified herein,

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of the inspection requirements specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification when such action is deemed necessary to assure supplies and services conform to perscribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.5 and all tests in 4.6, Failure of the first article tests shall be cause for rejection.

4.4 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Quality conformance inspection shall consist of the examination in 4.5 and the test in 4.6.1. Failure of the item to pass the examination or test shall be cause for rejection.

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4.5 Examination. The machine and equipment shall be examined to determine compliance with the requirements of 3.2 through 3.5 and 3.6 through 3.13.

4.6 Tests.

4.6.1 Operational test. The machine shall be operated at no load for not less than 60 minutes. proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period.

4.6.2 Alignment accuracy tests. Each machine shall be tested to determine that it complies with the alignment accuracy requirements of table II (see 3.7).

4.6.3 performance tests. Each shaper shall cut a 1 inch external keyway, 1/2 inch deep and 12 inches long in 25 minutes or less. The test cylinder shall be of SAE 1040 steel, 6 inch inside diameterby 8 inch outside diameter by 12 inches long. The tool holder and tool type used shall be manufacturer's standard tooling.

4.7 Packaging inspection. Packaging shall be inspected to determine compliance with the requirements of Section 5.

5. PACKAGING

5.1 Packaging - packing and markings. Unless otherwise specified, packaging, packing and marking shall be in accordance with ASTM Standard D-3951. When required, level A packaging and level A or B packing and marking shall be in accordance with the applicable requirements of MIL-M-81058. The required levels and special marking shall be as specified (see 6.2.1).

6. NOTES

6.1 Intended use. The shapers and slotters covered by this specification are intended for rough and finish machining of metal workplaces.

6.2 Ordering data.

6.2.1 Acquisition documents. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type, class, size, and style required (see 1.2).

. When a first article is required for inspection and approval (see 3.1, 4.3, and 6.3).

d. If machine is required to be configured in a specific measurement system (US or SI), state required system (see 3.2.1).

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Specify measurement system required (US or SI); if required, (see 3.2.1.1).

f. Dual calibrations (US and SI); if required, (see 3,2.1.2)."

g. Specify additional safety and health requirements (see 3.2.4).

h. Painting, if different (see 3.3.5).

i. Electrical system, if different (see 3.4.3).

j. Type motor enclosure, if different (see 3.4.3.1).

k. Type of ram drive (hydraulic or mechanical.) (see 3.4.7).

1. Standard equipment, if different (see 3.8).

m. Optional equipment, if required; fully describe (see 3.9).

n. Lubrication chart or plate, if different (see 3.10).

00 Nameplate, if different (see 3.11).

P" First article inspection, if different (see 4.3).

q. Quality conformance inspection, if different (see 4.4),

Specific packaging, packing and special marking, if required (see 5.1). (Specify level A packaging, specify level A or B packing and specify special marking, if required).

6.2.2 Contract data requirements. Required technical data, such as operator manuals, parts lists, wiring diagrams, and other instructions for operation and maintenance as identified on a numbered DD Form 1664 should be specified on DD Form 1423 incorporated into the contract.

6.3 First article. When a first article inspection is required, the end item will be tested and should be either a first article sample or a standard production sample from the contractor's current inventory. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination and test approval of the documents first article.

6.4 Safety and health requirements. Paragraph 3.2.4 requires compliance only with those OSHA requirements that concern the machine itself. It does not require compliance with those OSHA requirements that concern "the machine in its operating environment" such as noise levels, radiation levels, electromagnetic emissions, noxious vapors, air contaminants, heat, etc. Since OSHA limits the total hazard level of these hazards in the environment (and does not limit the hazard level of individual machines in the environment), the requesting activity is advised to analyze the existing hazard levels in the proposed operating environment and specify additional machine requirements

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that will integrate the new machine into its future operating environment. If specific point-of-operation guarding is required, the requesting activity should specify the exact configuration of the guard required as, in most cases, the guard configuration is dependent on the size and configuration of the workplaces. The above, and any other additional safety and health requirements, should be specified in detail under 6.2.1(g).

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - AL
Navy - SH
Air Force - 99

Review Activities:
Air Force - 84
DLA - GS

User Activities:
Army - SM
Navy - SH

Preparing Activity:
DLA-IP

Project Number:
3418-0026

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-80001C	Shapers, Metal Cutting, Vertical

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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